UCB Algorithm

- t : current round number
- $N_i(t)$: the number of times arm i was selected during rounds $1, \ldots, t$
- $\hat{\mu}_i(t) = \dfrac{\sum_{s=1}^{t} r_s \mathbf{1}\{A_s = i\}}{N_i(t)}$: estimated mean reward of arm i

Definition 1 (UCB algorithm)**.** *At the current round t,*

$$\mathrm{UCB}_i(t) = \hat{\mu}_i(t) + c\sqrt{\frac{\log t}{N_i(t)}} \tag{1}$$

The strategy that selects the arm i with the largest value, $A_t = \arg\max_i \mathrm{UCB}_i(t)$, is called the UCB algorithm.

Question. the natural Upper Confidence Bound for the mean reward μ_i appears to be $\boldsymbol{|\hat{\mu}_i(t) - \mu_i| = c\sqrt{\frac{1}{N_i(t)}}}$. (Sample variance)

But then why does the actual UCB algorithm scale the Upper Confidence Bound by $\log t$: $\boldsymbol{\mathrm{UCB}_i(t) = \hat{\mu}_i(t) + c\sqrt{\frac{\log t}{N_i(t)}}}$.

Derivation of UCB

Intuition

Without the $\log t$ correction, an Upper Confidence Bound algorithm can structurally **fail to perform the exploration** that is necessary.

The problem arises in the opposite case: when the intended role of **the Upper Confidence Bound fails**, so that it is no longer guaranteed that μ_i lies below the Upper Confidence Bound, and in fact the true mean reward μ_i **is larger than** the Upper Confidence Bound.

The quantity $\boldsymbol{\widetilde{\mathrm{UCB}}_i(t) := \hat{\mu}_i(t) + c\sqrt{\frac{1}{N_i(t)}}}$ **changes only when arm i is selected.**

Now consider the following situation. Suppose the optimal arm 1 has mean reward μ_1^{opt}, and another suboptimal arm 2 has mean reward $\mu_2^{\mathrm{sub}} < \mu_1^{\mathrm{opt}}$.

Assume that at some initial time t_0, because of an unfavorable sequence of observations r_t, the estimate $\widetilde{\mathrm{UCB}}_1(t_0)$ becomes **much smaller than** μ_1^{opt} and also μ_2^{sub}.

What happens then?

1. As arm 2 continues to be selected, $\widetilde{\mathrm{UCB}}_2(t)$ fluctuates mostly above μ_2^{sub} and converges to it.
2. Meanwhile, the severely underestimated $\widetilde{\mathrm{UCB}}_1(t_0)$ **is never selected.** Hence $\widetilde{\mathrm{UCB}}_1(t)$ **is never updated**, and it may happen that $\widetilde{\mathrm{UCB}}_1(t) < \widetilde{\mathrm{UCB}}_2(t)$ for all future times t.

As a consequence, this algorithm may keep playing the suboptimal arm **infinitely often**.

Preview

The heuristic of selecting the arm with the largest Upper Confidence Bound alone **was not sufficient to guarantee proper exploration,** and this is why the $\log t$ scaling is needed in UCB.

The goal of MAB is to choose a suboptimal wrong arm **only finitely** many times. (Otherwise, it incurs linear regret.)

The probability that the true μ_i is **larger than** $\mathrm{UCB}_i(t)$ is the **failure probability** of the confidence bound. This is also why arm 1 failed in the intuitive example above. If the true mean μ_i is lower than $\mathrm{UCB}_i(t)$, then the algorithm will identify the optimal arm.

However, since $\boldsymbol{\mathbb{P}(|\hat{\mu}_i(t) - \mu_i| > c\sqrt{\frac{1}{N_i(t)}}) = \text{const}}$ is independent of time, if arm i is **not selected**, this probability **cannot be reduced**.

1st Borel–Cantelli lemma

Meaning : if the accumulated sum of the probabilities that E_t occurs at each time t **is finite**, then it guarantees that E_t occurs **only finitely** many times almost surely.

Theorem 1 (1st Borel–Cantelli lemma)**.** *Here the event*

$$\limsup_{t\to\infty} E_t := \bigcap_{T=1}^{\infty} \bigcup_{t\geq T} E_t = \{E_t \ i.o.\} \tag{2}$$

is the event that E_t occurs infinitely often.

If some event E_t satisfies

$$\sum_{t=1}^{\infty} \mathbb{P}(E_t) < \infty \tag{3}$$

then

$$\mathbb{P}(\limsup_{t\to\infty} E_t) = \mathbb{P}(E_t \ i.o.) = 0 \tag{4}$$

That is, E_t occurs only finitely many times almost surely.

Hoeffding's inequality

Meaning : the **deviation** (upper tail) from the mean is **bounded by** $P(\hat{\mu}_i(t) - \mu_i \geq \epsilon_i(t)) \leq e^{-2N_i(t)\epsilon_i(t)^2}$.

Theorem 2 (Hoeffding's inequality)**.** *For independent random variables $X_1, \ldots, X_n$, if each X_i is σ_i^2-subgaussian, then it is known that $S_n := \sum_{i=1}^n (X_i - \mathbb{E}[X_i])$ is $\sum_{i=1}^n \sigma_i^2$-subgaussian.*

Therefore, for all $n \in \mathbb{N}$, $\epsilon > 0$,

$$\mathbb{P}(S_n \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sum_{i=1}^n \sigma_i^2}\right) \tag{5}$$

$$\mathbb{P}(\frac{S_n}{n} \geq \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right) \tag{6}$$

$$= \exp\left(-2n\epsilon^2\right), \quad \textit{when } X_i \in [0,1] \textit{ so } \sigma^2 = \frac{1}{4} \tag{7}$$

Usually, σ^2 is a separately specified constant. For example, if $X_i \in [0,1]$ is a random variable, then $\sigma^2 = \frac{1}{4}$. Therefore, when σ^2 is not written explicitly, it is usually understood in this way.

Proof

If we use it without the $\log t$ correction, $Pr(\text{Upper Confidence Bound fails}) = \boldsymbol{P(\hat{\mu}_i(t) - \mu_i \geq c\sqrt{\frac{1}{N_i(t)}}) \leq e^{-2c^2} = \text{const}}$ (Hoeffding's inequality). Its infinite sum diverges.

That is, there is no guarantee that $\boldsymbol{P(\hat{\mu}_i(t) - \mu_i \geq c\sqrt{\frac{1}{N_i(t)}})}$ happens only finitely many times (Borel–Cantelli lemma), and in bad cases it may happen infinitely many times.

Therefore, in order for the infinite sum of the probabilities at each time t to converge, it would be appropriate to have $\boldsymbol{P(\hat{\mu}_i(t) - \mu_i \geq \epsilon_i(t)) \leq e^{-2N_i(t)\epsilon_i(t)^2} \asymp t^{-\alpha}}$, $\alpha > 1$. This is because $\boldsymbol{\sum_{t=1}^{\infty} \frac{1}{t^\alpha} < \infty}$.

(Here $A_t \asymp B_t$ means $A_t = \Theta(B_t)$, that is, $\exists c_1, c_2 > 0\ \forall n \in \mathbb{N} : c_1 B_t \leq A_t \leq c_2 B_t$. It means that the two terms have the same order.)

Therefore

$$e^{-2N_i(t)\epsilon_i(t)^2} \asymp t^{-\alpha} \tag{8}$$

$$-2N_i(t)\epsilon_i(t)^2 \asymp -\alpha \log t \tag{9}$$

$$\boldsymbol{\epsilon_i(t) \asymp \sqrt{\frac{\log t}{N_i(t)}}} \tag{10}$$

is necessary.

That is, if $\epsilon_i(t)$ is constructed only in terms of the **number of selections $\boldsymbol{N_i(t)}$** and not time, as $c\sqrt{\frac{1}{N_i(t)}}$, then we cannot surely prevent the Upper Confidence Bound from **failing infinitely** many times.

Conclusion of UCB

In conclusion, compared to intuitive understanding, we end up **exploring more** by the amount of the correction due to $\log t$.

The reason is as follows. If we use the Upper Confidence Bound as it is, then we may **never select** an arm i such that $\mu_i \geq \hat{\mu}_i(t) + c\sqrt{\frac{1}{N_i(t)}}$.

As a result, there arises a probability that we **keep playing a non-optimal arm infinitely often** by mistake.

Therefore, only when we scale the Upper Confidence Bounds of all arms by $\log t$, so that the $\text{UCB}_i(t)$ of arms that are not being selected **gradually increases** and **exploration increases**, can we guarantee that there is no possibility of choosing a **non-optimal wrong arm infinitely often**.